RESULTS

SELF-X algorithm comparison

SELF-X Agentic averaged **60.54 ms delay**, **24.88% loss** and **6.62 Mbps goodput** across 12 paired held-out traces.

Against DQN, Agentic delay reduction was +1.22% and loss reduction was -0.33 pp. Against DPP alone, the delay difference was +0.12 ms (Holm p = 0.0745).

Comparator	Delay (ms)	Delay reduction	Paired 95% CI (ms)	Holm p	Loss reduction (pp)	Goodput gain
Static	80.88	+25.2%	[-27.51, -13.18]	0.0102	+1.12	+1.7%
Base LM	61.30	+1.2%	[-0.96, -0.55]	0.00143	-0.33	-0.4%
Base LM + critique	61.34	+1.3%	[-1.00, -0.61]	0.00119	-0.33	-0.4%
LoRA policy	79.40	+23.8%	[-26.01, -11.70]	0.0102	+1.06	+1.6%
Agentic at 5 s	71.77	+15.6%	[-14.57, -7.90]	0.0048	+0.45	+0.7%
Capacity threshold	69.04	+12.3%	[-10.70, -6.30]	0.00143	+0.34	+0.6%
Least loaded	70.85	+14.6%	[-13.67, -6.95]	0.00704	+0.41	+0.7%
Tabular Q	71.52	+15.4%	[-14.11, -7.86]	0.00704	+0.50	+0.8%
DQN	61.29	+1.2%	[-0.96, -0.54]	0.00143	-0.33	-0.4%
Two-agent IQL	61.29	+1.2%	[-0.96, -0.54]	0.00143	-0.33	-0.4%
DPP only	60.42	-0.2%	[+0.01, +0.23]	0.0745	-0.31	-0.4%
Agentic without LoRA	60.07	-0.8%	[+0.34, +0.61]	0.00537	-0.33	-0.4%
Agentic without governor	60.41	-0.2%	[+0.00, +0.27]	0.0745	-0.30	-0.4%

Positive reductions and gains favour Agentic. The confidence interval is for Agentic delay minus comparator delay, so negative values favour Agentic. Intervals are pointwise. Holm p-values adjust the 13 two-sided delay comparisons. Loss reduction is in percentage points.

Which components actually ran?

Matched traces. Four seeds per profile for steady bursts, flash crowds and volatile traffic. Each controller saw identical demand per seed on two finite-buffer 10 Mbps paths. All actions went through the same guard. The evaluation models traffic in bits and delay in fluid slots.

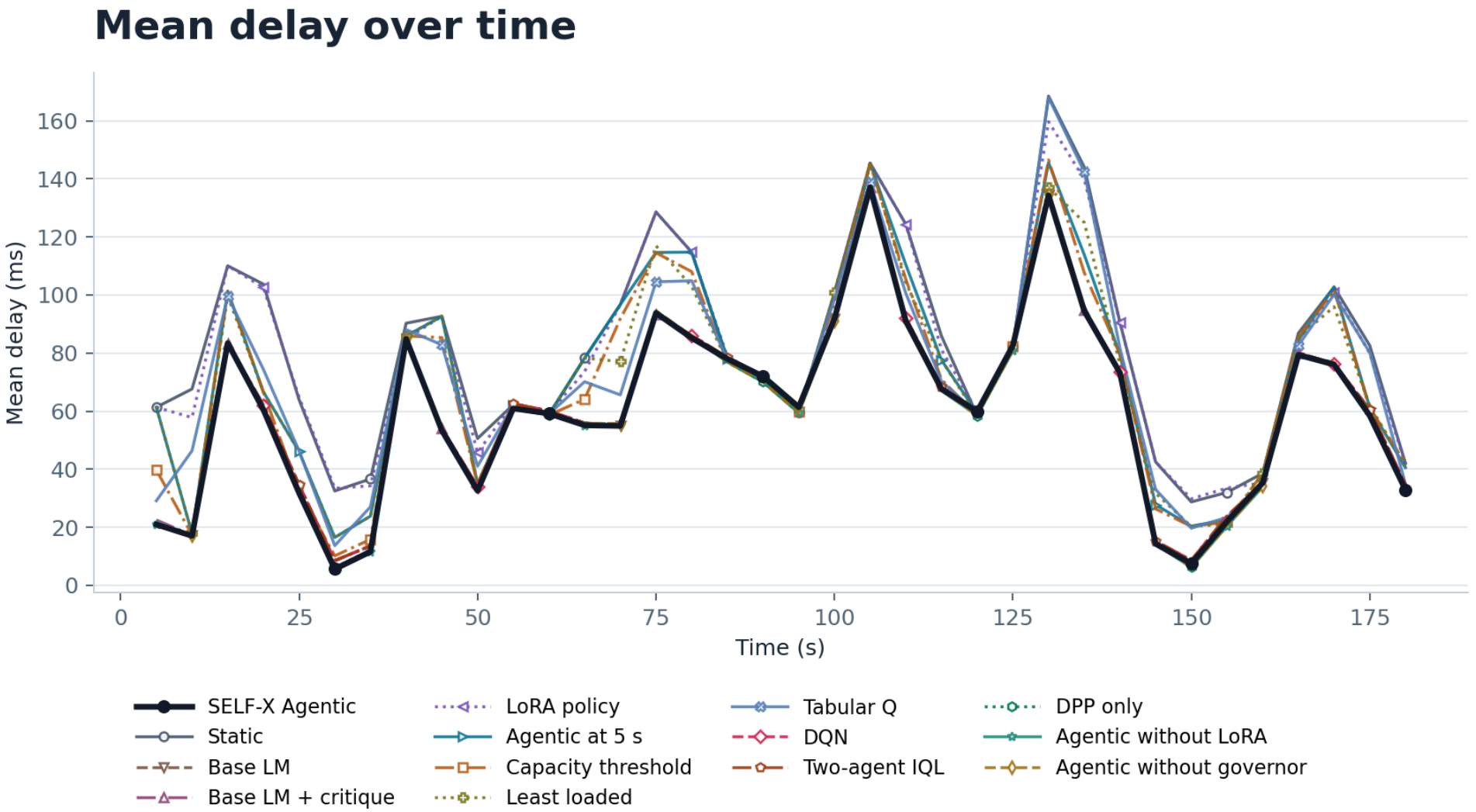
DQN and two-agent independent Q-learning. DQN ran 2,692 gradient steps. Routing and metering agents ran 2,692 each, with separate networks and replay buffers and a shared reward. Neural policies were frozen and greedy at test time. The guard reduced their different proposals to identical effective action sequences. The MARL baseline is project-specific.

Critique and adaptation. S2 invoked critique 432 times, used nonempty outcome memory 420 times and revised 0 proposals. S3 used the shipped SmolLM2 adapter. The full stack uses its output as a bounded DPP action preference.

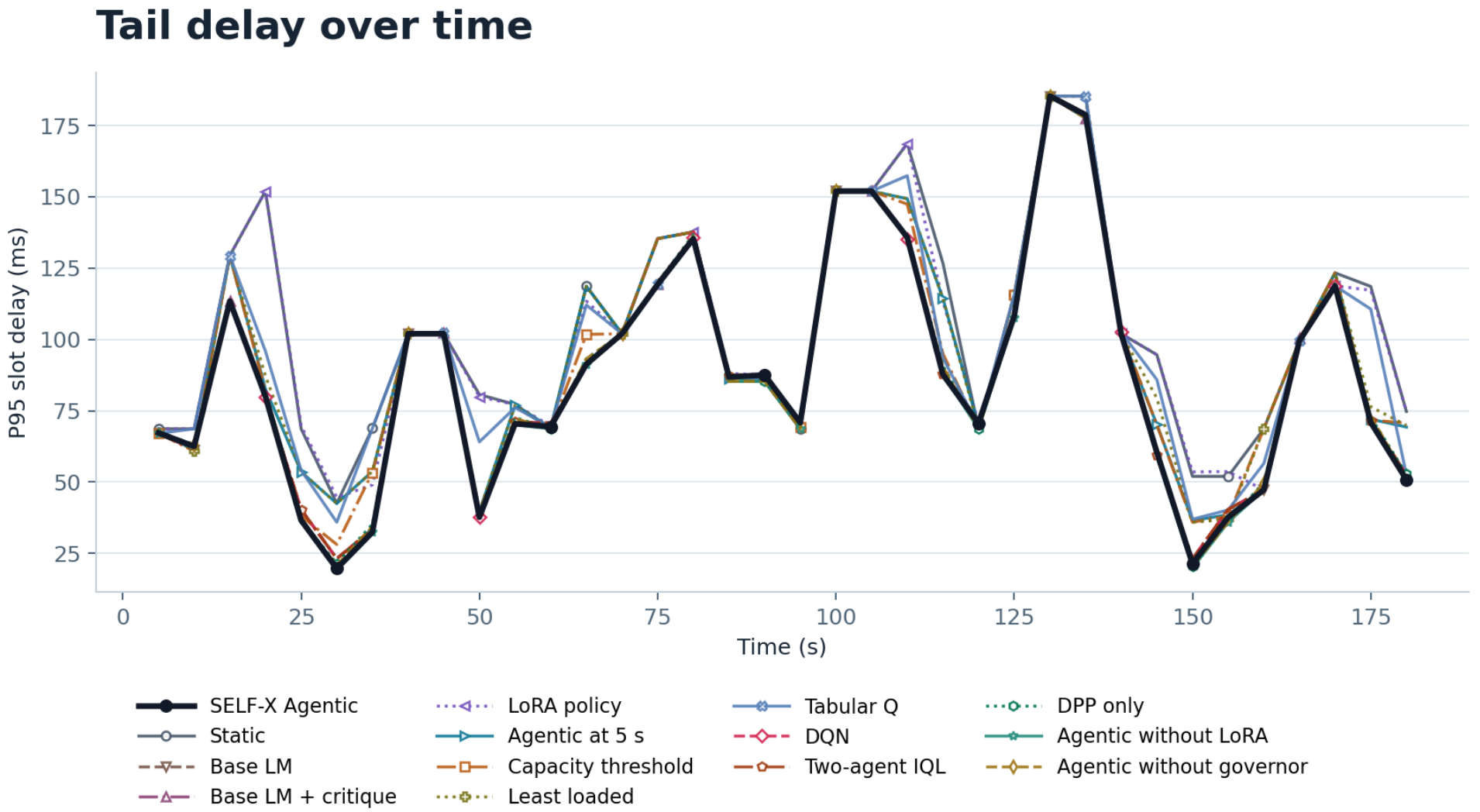
Governance and ablations. The full stack requested 72 orchestrator decisions and 360 role ballots from one shared local model. The orchestrator chose the same adjustment at every epoch. It changed the objective 72 times. The LoRA prior changed 2,920 of 21,600 optimiser choices. Separate arms disable each layer, disable both, or run the same optimisation every 5 s.

Timing and inference. Fast controllers saw current slot observations every 100 ms. Slow controllers sampled every 5 s. Each local model pass was assigned 400 ms of simulated delay. Specialists formed one parallel batch followed by the orchestrator. Prompt results were memoised. The timing budget is an experimental assumption.

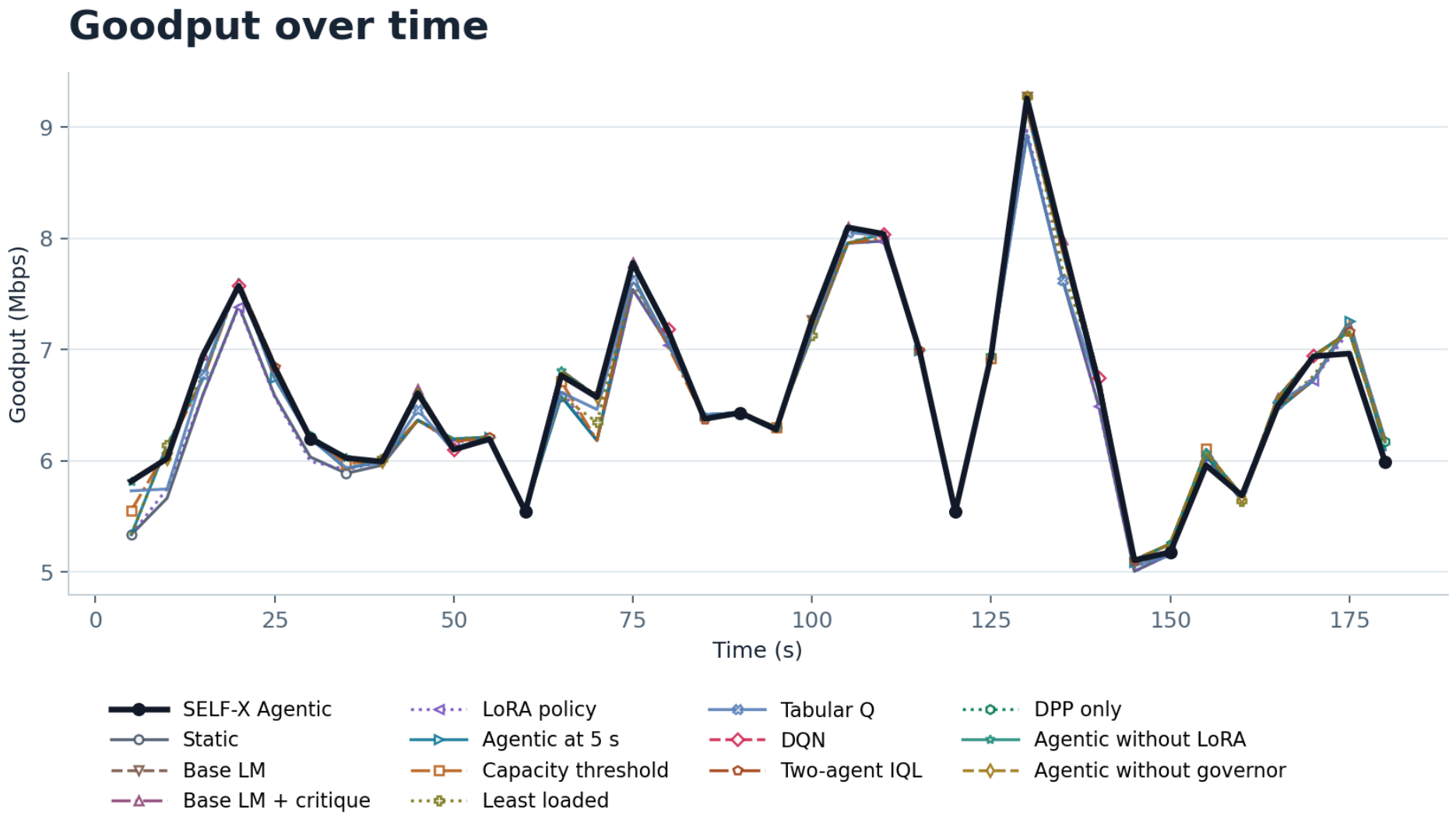
Scope. The paired tests estimate differences over traffic seeds for these trained policies and this topology. The study does not measure hardware performance or establish novelty. Earlier proxy-based comparisons are superseded.



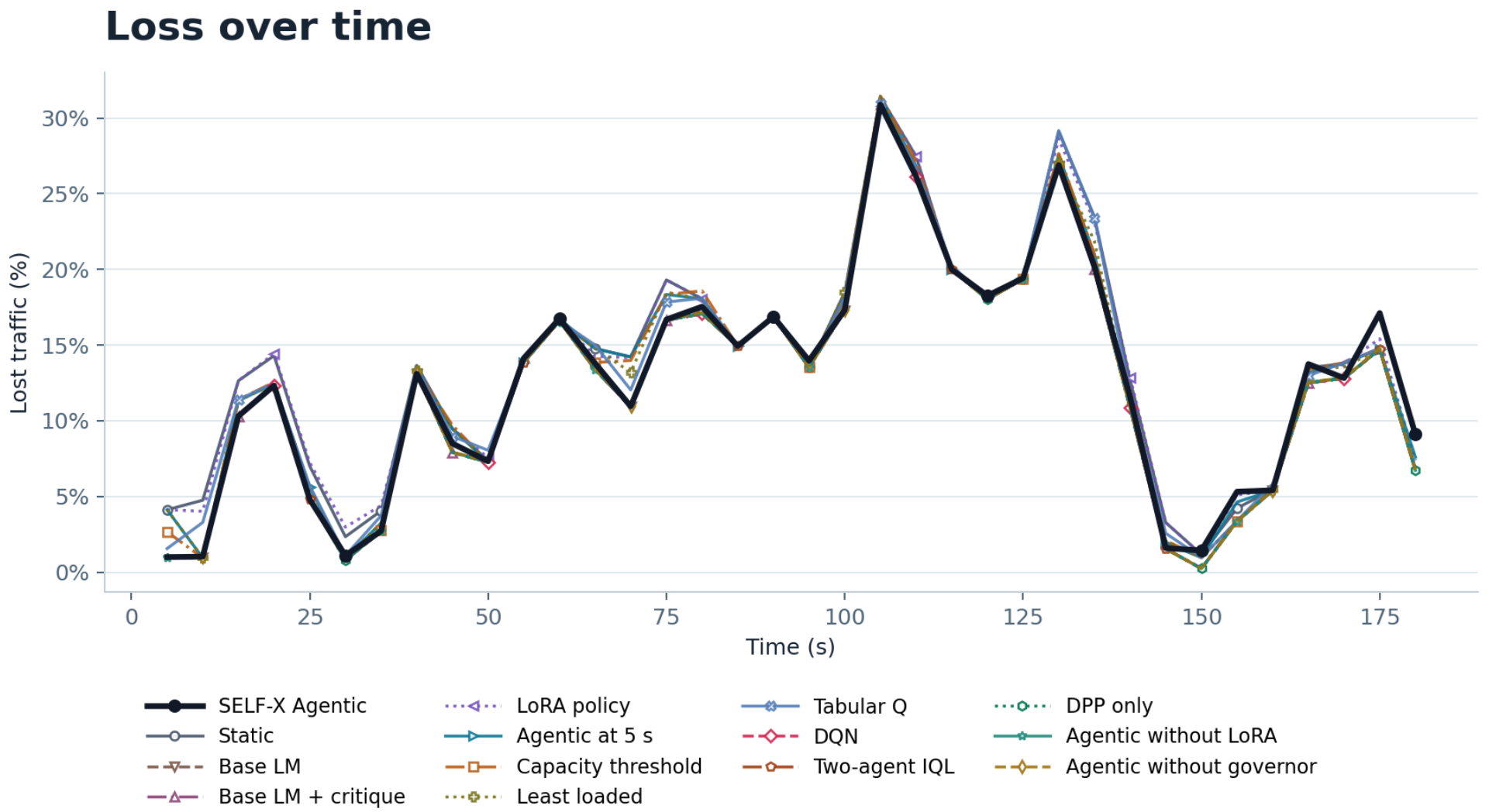
Means across paired test seeds at each 5 s window. Overlapping curves can share a position.



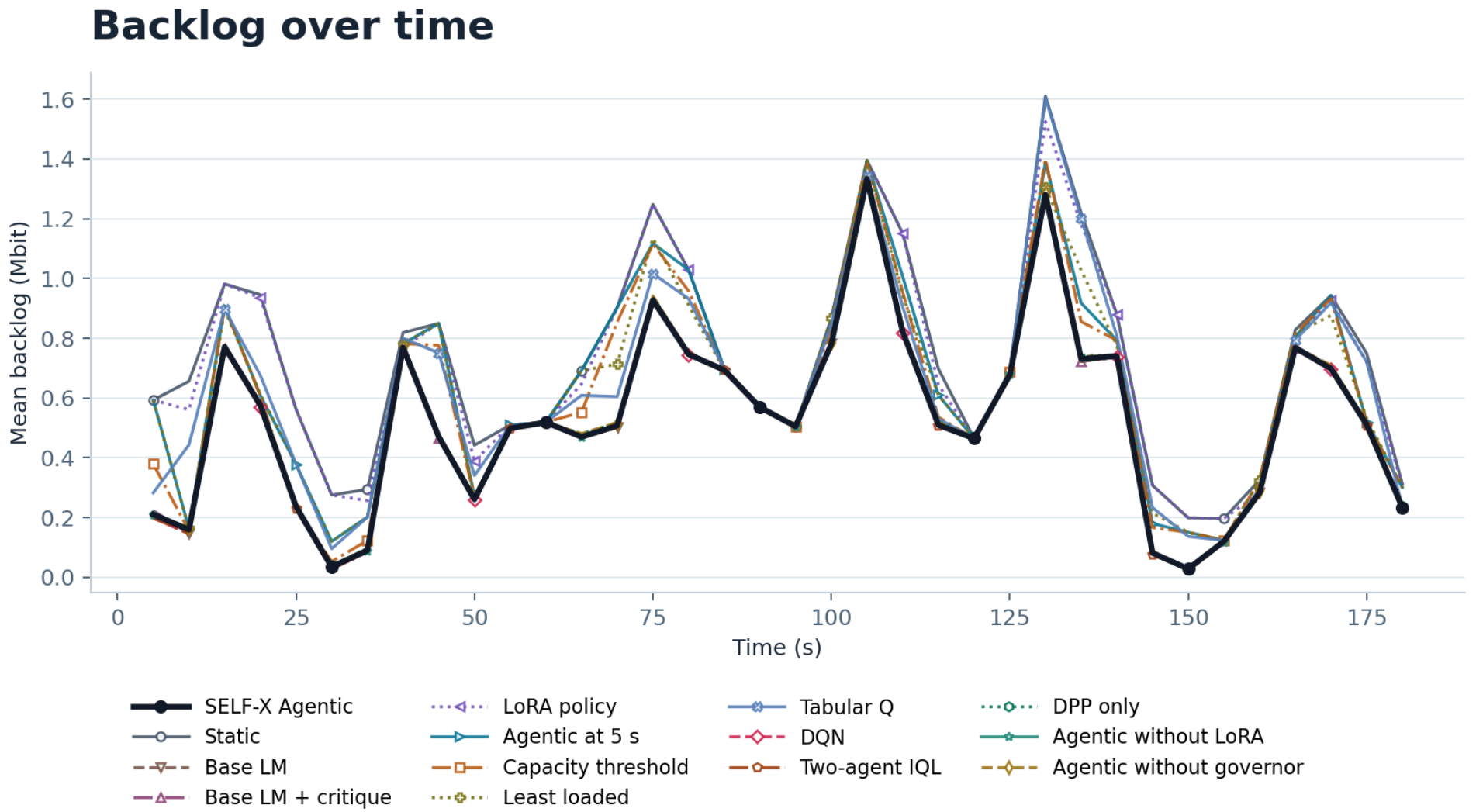
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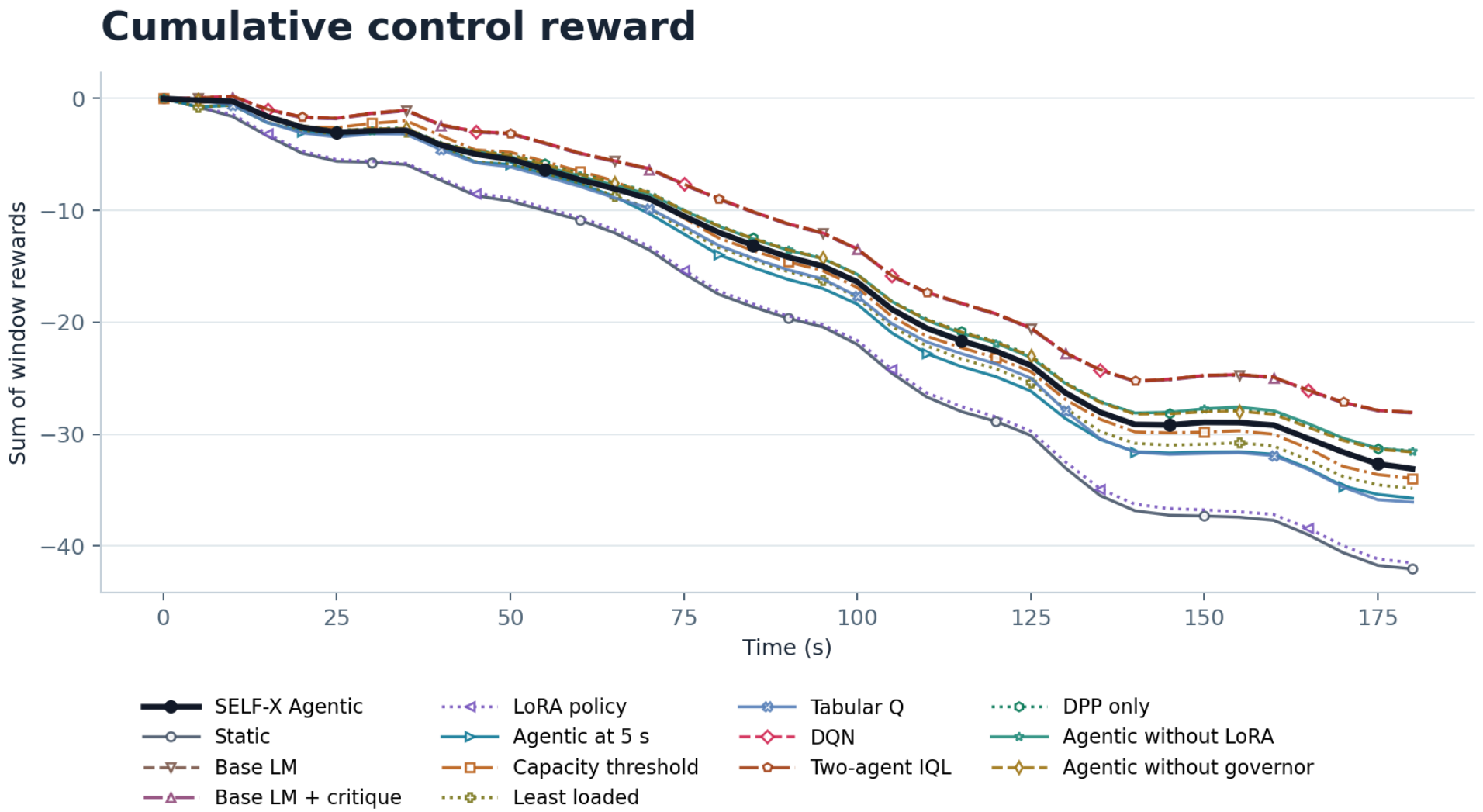
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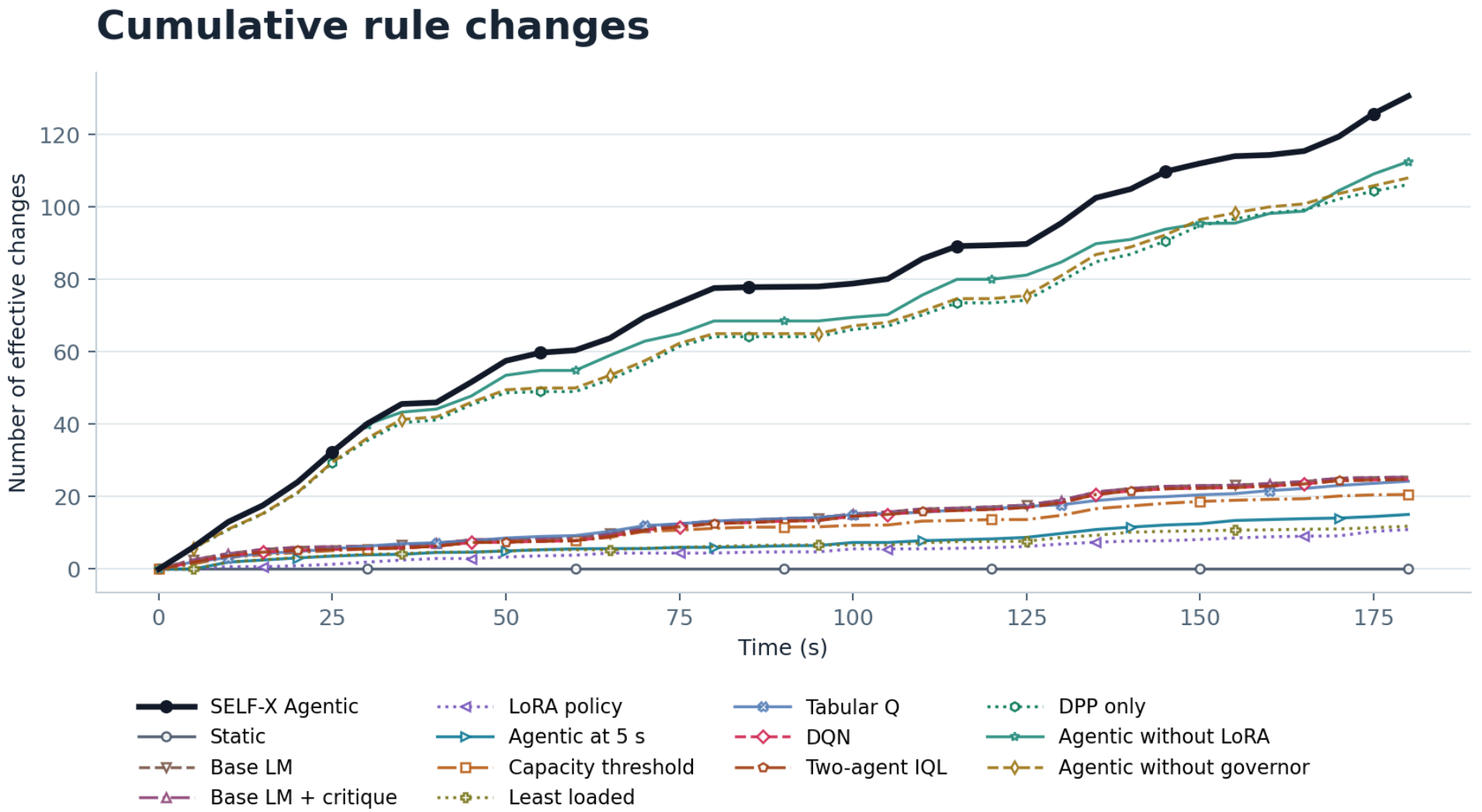
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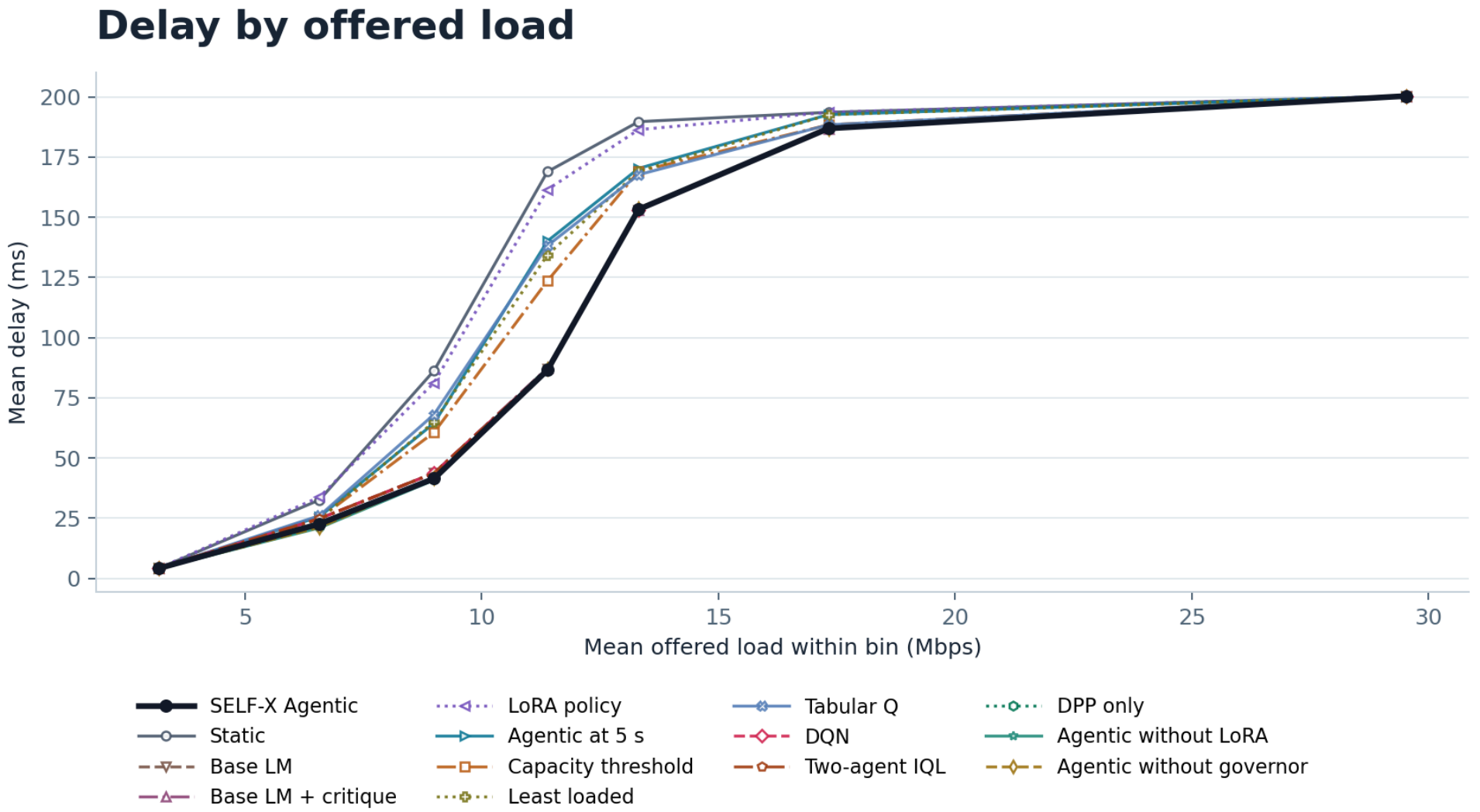
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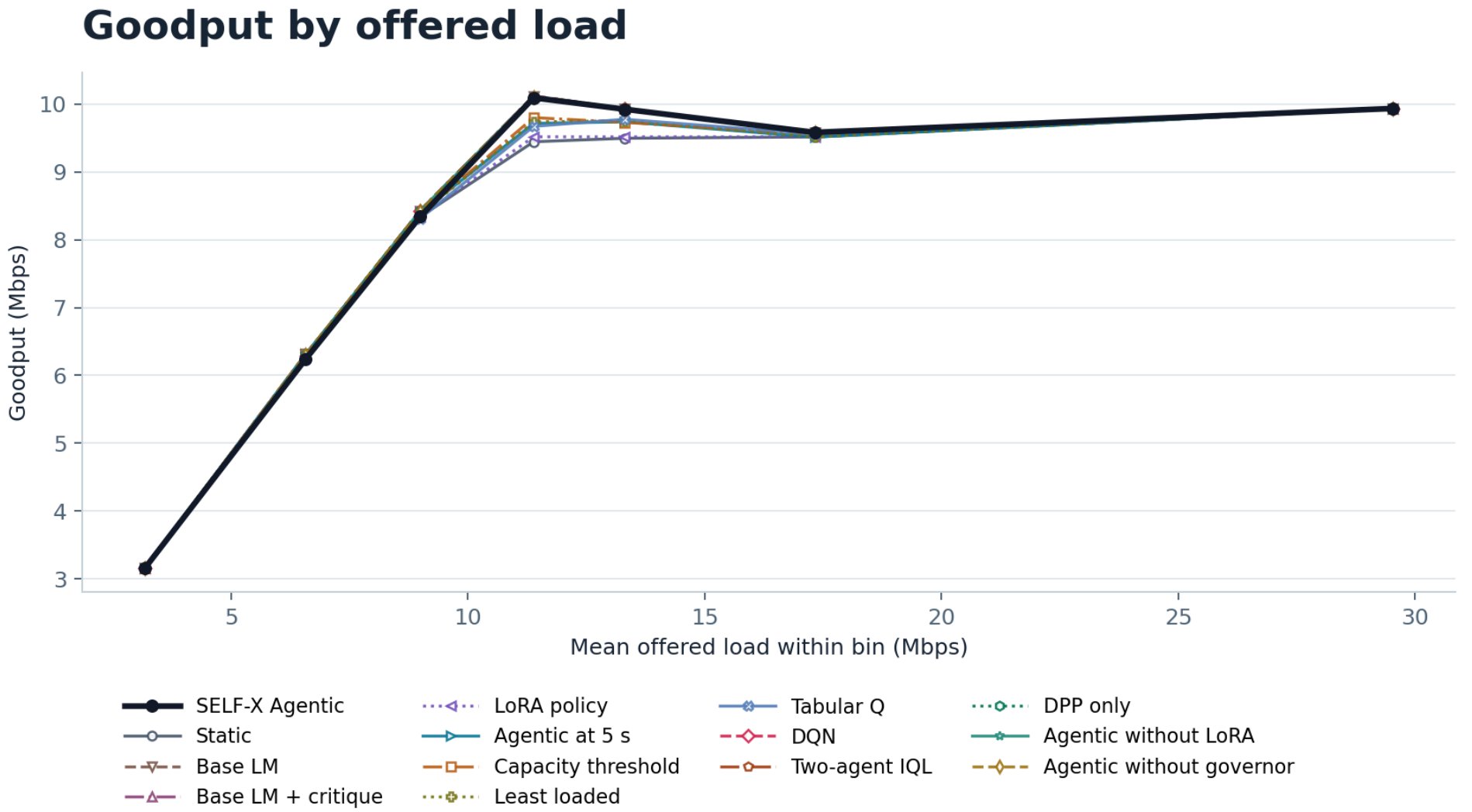
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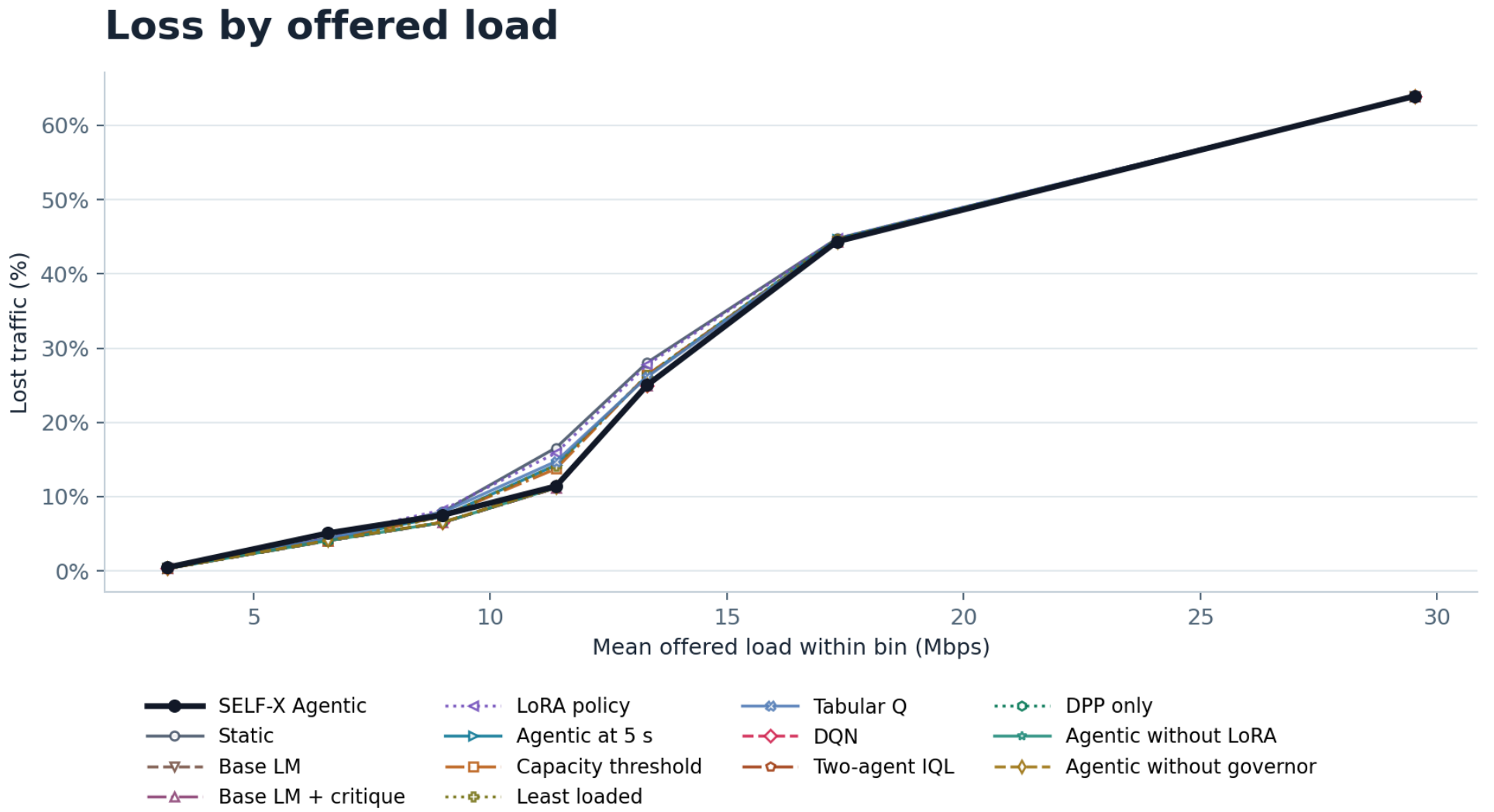
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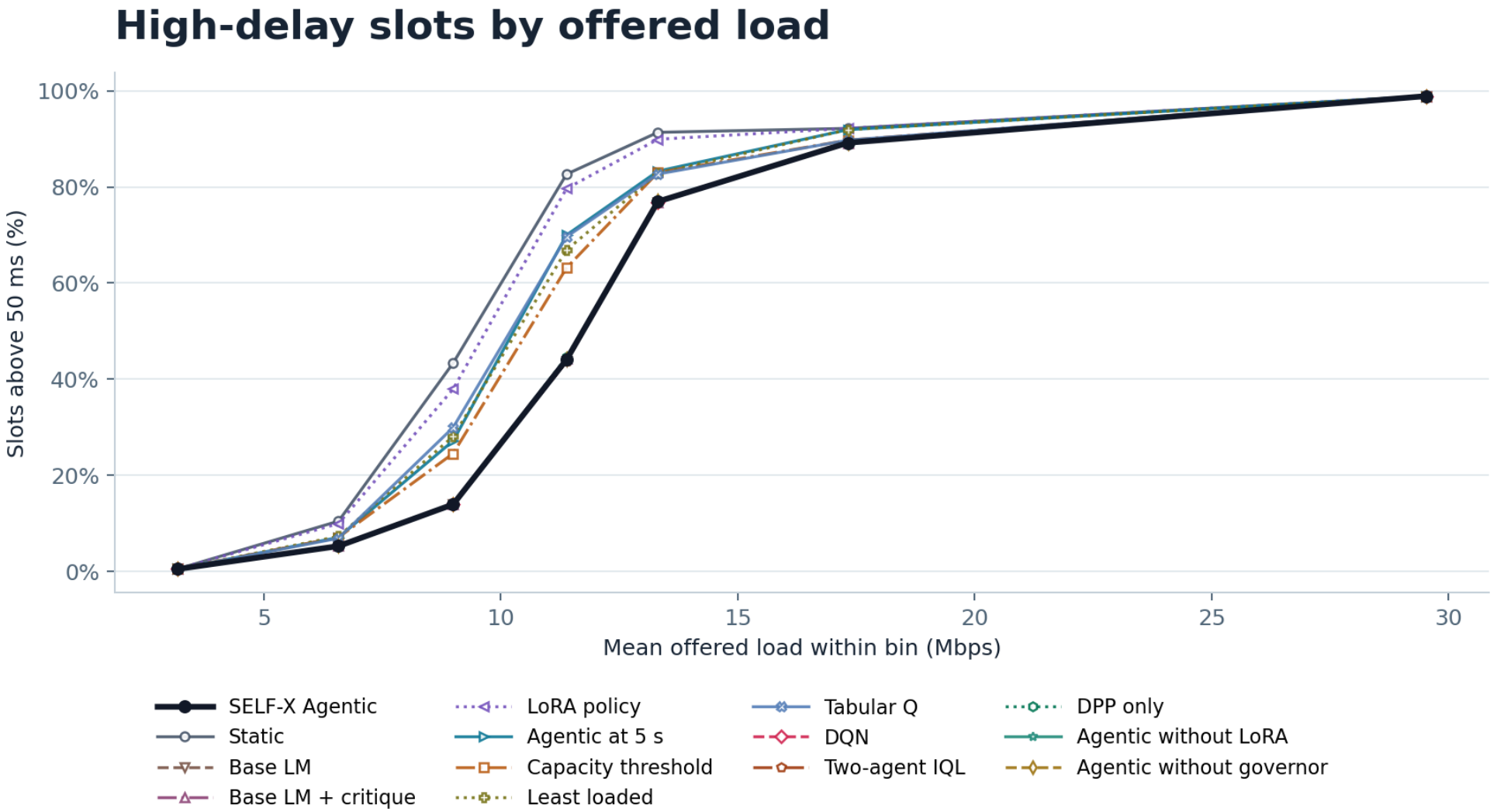
Common demand bins with at least eight windows. Left-to-right counts per controller: 156, 82, 44, 49, 50, 22, 29.



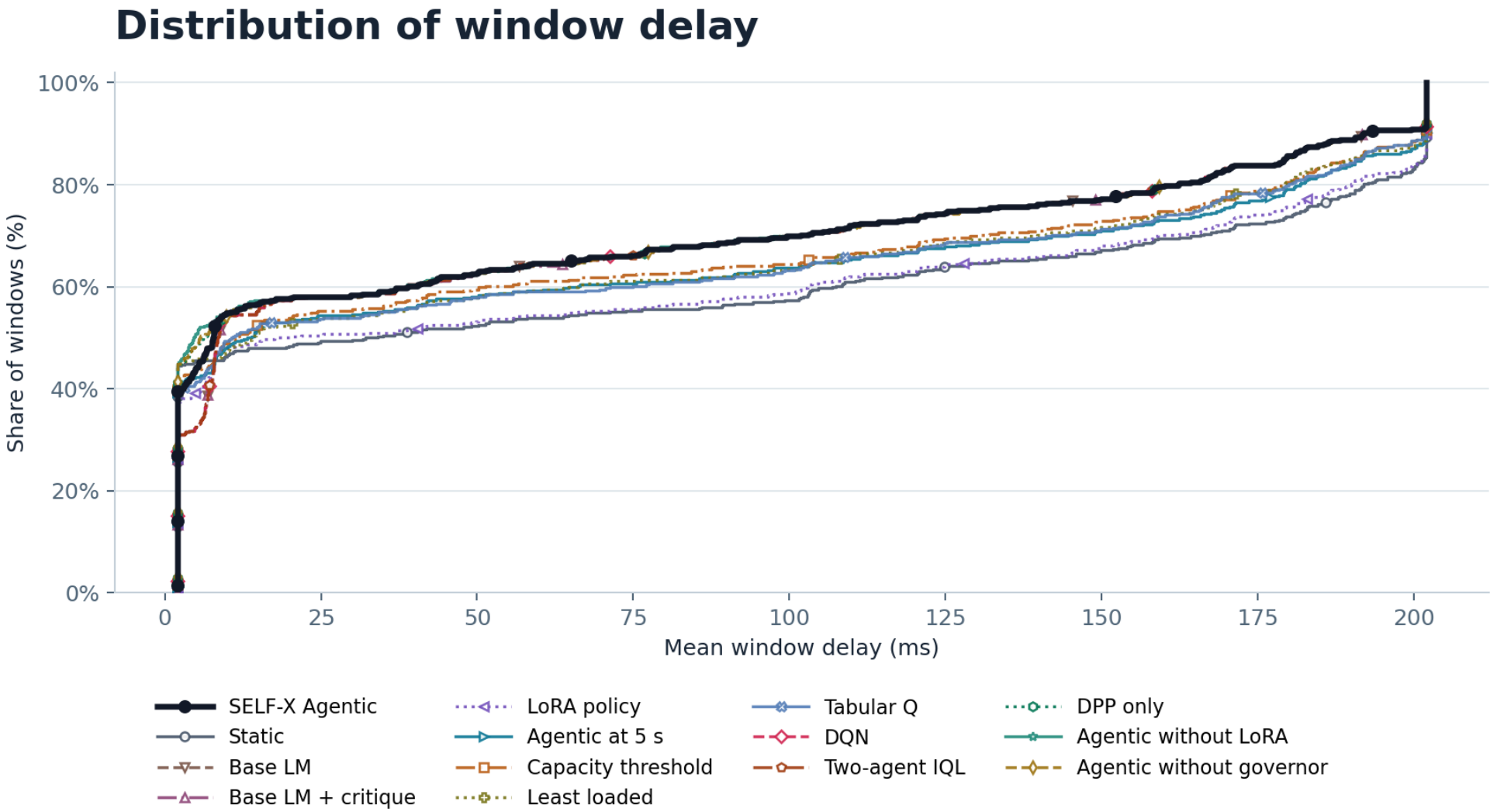
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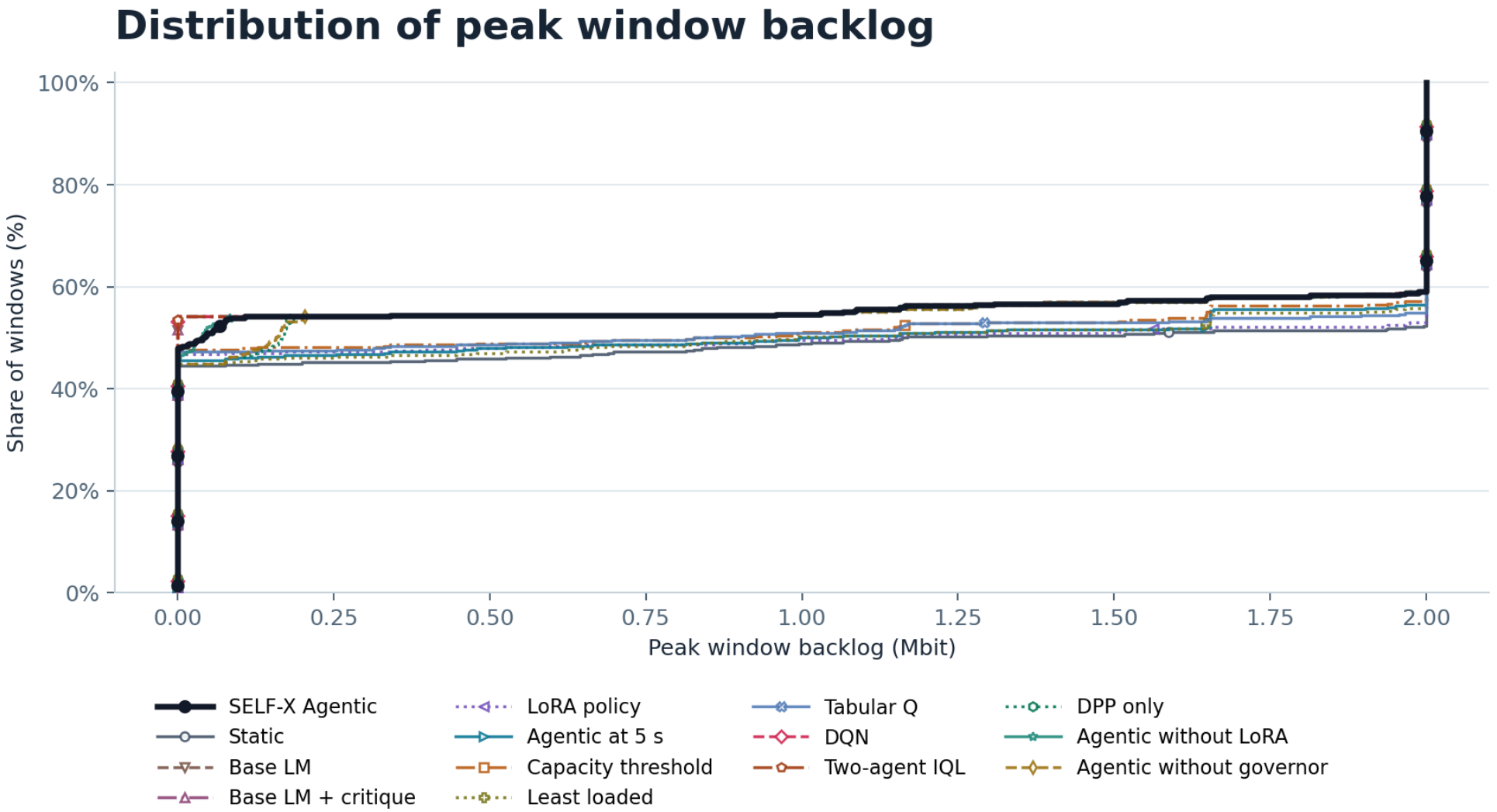
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Empirical distribution of 5 s windows pooled across the paired test traces.



Empirical distribution of 5 s windows pooled across the paired test traces.